```
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> SHOW DATABASES;
+--------------------------+
| Database                 |
+--------------------------+
| information_schema       |
| mysql                    |
| online_proctoring_system |
| performance_schema       |
| sakila                   |
| sys                      |
| world                    |
+--------------------------+
7 rows in set (0.08 sec)

mysql> USE ONLINE_PROCTORING_SYSTEM;
Database changed
mysql> SHOW TABLES;
+-----------------------------------+
| Tables_in_online_proctoring_system |
+-----------------------------------+
| exams                             |
| proctor_sessions                  |
| user_sessions                     |
+-----------------------------------+
3 rows in set (0.02 sec)

mysql> SELECT*FROM EXAMS;
+---------+-------------------------------+------------------+-------------+------------+
| EXAM_ID | EXAM_TITLE                    | DURATION_MINUTES | TOTAL_MARKS | PASS_MARKS |
+---------+-------------------------------+------------------+-------------+------------+
|     101 | PYTHON CERTIFICATION          |              100 |         100 |         50 |
|     102 | JAVA CERTIFICATION            |              120 |         100 |         55 |
|     103 | C PROGRAMMING CERTIFICATION   |               90 |         100 |         50 |
|     104 | C++ PROGRAMMING CERTIFICATION |              100 |         100 |         50 |
|     105 | DATA STRUCTURES               |              120 |         100 |         60 |
|     106 | DBMS CERTIFICATION            |               90 |         100 |         50 |
|     107 | OS CERTIFICATION              |               90 |         100 |         50 |
```

```
mysql> SELECT*FROM PROCTOR_SESSIONS;
+--------------------+---------+------------+-----------------+---------------------+---------------------+
| PROCTOR_SESSION_ID | EXAM_ID | SESSION_ID | PROCTOR_NAME    | SESSION_START       | SESSION_END         |
+--------------------+---------+------------+-----------------+---------------------+---------------------+
|                501 |     101 |          1 | Proctor Ravi    | 2026-01-04 09:00:00 | 2026-01-04 10:30:00 |
|                503 |     103 |          3 | Proctor Karthik | 2026-01-04 09:30:00 | 2026-01-04 10:45:00 |
|                504 |     104 |          4 | Proctor Divya   | 2026-01-04 10:00:00 | 2026-01-04 11:30:00 |
|                505 |     105 |          5 | Proctor Suresh  | 2026-01-04 10:15:00 | NULL                |
|                506 |     106 |          6 | Proctor Meena   | 2026-01-04 11:00:00 | 2026-01-04 12:30:00 |
|                507 |     107 |          7 | Proctor Arjun   | 2026-01-04 11:15:00 | 2026-01-04 13:15:00 |
+--------------------+---------+------------+-----------------+---------------------+---------------------+
6 rows in set (0.00 sec)

mysql> SELECT*FROM USER_SESSIONS;
+------------+---------+---------------------+---------------------+-------------+
| SESSION_ID | USER_ID | LOGIN_TIME          | LOGOUT_TIME         | IP_ADDRESS  |
+------------+---------+---------------------+---------------------+-------------+
|          1 |     201 | 2026-01-04 09:00:00 | 2026-01-04 10:30:00 | 192.168.1.1 |
|          3 |     203 | 2026-01-04 09:30:00 | 2026-01-04 10:45:00 | 192.168.1.3 |
|          4 |     204 | 2026-01-04 10:00:00 | 2026-01-04 11:30:00 | 192.168.1.4 |
|          5 |     205 | 2026-01-04 10:15:00 | NULL                | 192.168.1.5 |
|          6 |     206 | 2026-01-04 11:00:00 | 2026-01-04 12:30:00 | 192.168.1.6 |
|          7 |     207 | 2026-01-04 11:15:00 | 2026-01-04 13:15:00 | 192.168.1.7 |
|          8 |     208 | 2026-01-04 11:30:00 | NULL                | 192.168.1.8 |
+------------+---------+---------------------+---------------------+-------------+
7 rows in set (0.01 sec)

mysql> ALTER TABLE USER_SESSIONS ADD DEVICE_TYPE VARCHAR(50);
Query OK, 0 rows affected (0.11 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> ALTER TABLE USER_SESSIONS MODIFY EXAM_TITLE VARCHAR(50);
ERROR 1054 (42S22): Unknown column 'EXAM_TITLE' in 'user_sessions'
mysql> ALTER TABLE USER_SESSIONS MODIFY IP_ADDRESS VARCHAR(50);
Query OK, 0 rows affected (0.03 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

```
mysql> ALTER TABLE USER_SESSIONS MODIFY EXAM_TITLE VARCHAR(50);
ERROR 1054 (42S22): Unknown column 'EXAM_TITLE' in 'user_sessions'
mysql> ALTER TABLE USER_SESSIONS MODIFY IP_ADDRESS VARCHAR(50);
Query OK, 0 rows affected (0.03 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> ALTER TABLE USER_SESSIONS DROP DEVICE_TYPE;
Query OK, 0 rows affected (0.04 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> RENAME TABLE EXAMS TO CERTIFICATION_EXAMS;
Query OK, 0 rows affected (0.08 sec)

mysql> RENAME TABLE CERTIFICATION_EXAMS TO EXAMS;
Query OK, 0 rows affected (0.04 sec)

mysql> SELECT*FROM USER_SESSIONS WHERE LOGOUT_TIME IS NULL;
+------------+---------+---------------------+-------------+-------------+
| SESSION_ID | USER_ID | LOGIN_TIME          | LOGOUT_TIME | IP_ADDRESS  |
+------------+---------+---------------------+-------------+-------------+
|          5 |     205 | 2026-01-04 10:15:00 | NULL        | 192.168.1.5 |
|          8 |     208 | 2026-01-04 11:30:00 | NULL        | 192.168.1.8 |
+------------+---------+---------------------+-------------+-------------+
2 rows in set (0.00 sec)

mysql> UPDATE EXAMS SET DURATION_MINUTES =110 WHERE EXAM_ID =105;
Query OK, 1 row affected (0.02 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> DELETE FROM PROCTOR_SESSIONS WHERE SESSION_ID =5;
Query OK, 1 row affected (0.01 sec)

mysql> SELECT*FROM EXAMS ORDER BY DURATION_MINUTES DESC;
+---------+-------------------------------+------------------+-------------+------------+
| EXAM_ID | EXAM_TITLE                    | DURATION_MINUTES | TOTAL_MARKS | PASS_MARKS |
+---------+-------------------------------+------------------+-------------+------------+
|     102 | JAVA CERTIFICATION            |              120 |         100 |         55 |
|     105 | DATA STRUCTURES               |              110 |         100 |         60 |
|     101 | PYTHON CERTIFICATION          |              100 |         100 |         50 |
|     104 | C++ PROGRAMMING CERTIFICATION |              100 |         100 |         50 |
```

```
mysql> SELECT PROCTOR_NAME,COUNT(*) AS TOTAL_SESSIONS FROM PROCTOR_SESSIONS GROUP BY PROCTOR_NAME;
+----------------+----------------+
| PROCTOR_NAME   | TOTAL_SESSIONS |
+----------------+----------------+
| Proctor Ravi   |              1 |
| Proctor Karthik |              1 |
| Proctor Divya  |              1 |
| Proctor Meena  |              1 |
| Proctor Arjun  |              1 |
+----------------+----------------+
5 rows in set (0.01 sec)

mysql> SELECT E.EXAM_TITLE,P.PROCTOR_NAME FROM EXAMS E JOIN PROCTOR_SESSIONS P ON E.EXAM_ID =P.EXAM_ID;
+------------------------------+----------------+
| EXAM_TITLE                   | PROCTOR_NAME   |
+------------------------------+----------------+
| PYTHON CERTIFICATION         | Proctor Ravi   |
| C PROGRAMMING CERTIFICATION  | Proctor Karthik |
| C++ PROGRAMMING CERTIFICATION | Proctor Divya  |
| DBMS CERTIFICATION           | Proctor Meena  |
| OS CERTIFICATION             | Proctor Arjun  |
+------------------------------+----------------+
5 rows in set (0.00 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> SAVEPOINT SP1;
Query OK, 0 rows affected (0.00 sec)

mysql> ROLLBACK;
Query OK, 0 rows affected (0.00 sec)

mysql> COMMIT;
```

```
mysql> SELECT COUNT(*) FROM TOTAL_EXAMS FROM EXAMS;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to
ROM EXAMS' at line 1
mysql> SELECT COUNT(*) AS TOTAL_EXAMS FROM EXAMS;
+-------------+
| TOTAL_EXAMS |
+-------------+
|           7 |
+-------------+
1 row in set (0.01 sec)

mysql> SELECT SUM(TOTAL_MARKS) AS TOTAL_MARKS_SUM FROM EXAMS;
+-----------------+
| TOTAL_MARKS_SUM |
+-----------------+
|             700 |
+-----------------+
1 row in set (0.01 sec)

mysql> SELECT AVG (DURATION_MINUTES) AS AVG_DURATION FROM EXAMS;
+--------------+
| AVG_DURATION |
+--------------+
|     100.0000 |
+--------------+
1 row in set (0.00 sec)

mysql> SELECT MIN(DURATION_MINUTES) AS MIN_DURATION FROM EXAMS;
+--------------+
| MIN_DURATION |
+--------------+
|           90 |
+--------------+
1 row in set (0.00 sec)

mysql> SELECT MAX(DURATION_MINUTES) AS MAX_DURATION FROM EXAMS;
+--------------+
| MAX_DURATION |
```

```
mysql> CREATE VIEW EXAM_DETAILS AS SELECT EXAM_ID,EXAM_TITLE,DURATION_MINUTES,PASS_MARKS FROM EXAMS;
Query OK, 0 rows affected (0.02 sec)

mysql> SELECT * FROM EXAM_DETAILS;
+---------+-----------------------------+------------------+------------+
| EXAM_ID | EXAM_TITLE                  | DURATION_MINUTES | PASS_MARKS |
+---------+-----------------------------+------------------+------------+
|     101 | PYTHON CERTIFICATION        |              100 |         50 |
|     102 | JAVA CERTIFICATION          |              120 |         55 |
|     103 | C PROGRAMMING CERTIFICATION |               90 |         50 |
|     104 | C++ PROGRAMMING CERTIFICATION |              100 |         50 |
|     105 | DATA STRUCTURES             |              110 |         60 |
|     106 | DBMS CERTIFICATION          |               90 |         50 |
|     107 | OS CERTIFICATION            |               90 |         50 |
+---------+-----------------------------+------------------+------------+
7 rows in set (0.00 sec)

mysql> SELECT e.exam_title, p.proctor_name FROM Exams e INNER JOIN Proctor_Sessions p ON e.exam_id = p.exam_id;
+-----------------------------+----------------+
| exam_title                  | proctor_name   |
+-----------------------------+----------------+
| PYTHON CERTIFICATION        | Proctor Ravi   |
| C PROGRAMMING CERTIFICATION | Proctor Karthik |
| C++ PROGRAMMING CERTIFICATION | Proctor Divya  |
| DBMS CERTIFICATION          | Proctor Meena  |
| OS CERTIFICATION            | Proctor Arjun  |
+-----------------------------+----------------+
5 rows in set (0.00 sec)

mysql> SELECT e.exam_title, p.proctor_name FROM Exams e LEFT JOIN Proctor_Sessions p ON e.exam_id = p.exam_id;
+-----------------------------+----------------+
| exam_title                  | proctor_name   |
+-----------------------------+----------------+
| PYTHON CERTIFICATION        | Proctor Ravi   |
| JAVA CERTIFICATION          | NULL           |
| C PROGRAMMING CERTIFICATION | Proctor Karthik |
| C++ PROGRAMMING CERTIFICATION | Proctor Divya  |
| DATA STRUCTURES             | NULL           |
```